

PCA-Assisted LSTM Autoencoders for Anomaly Detection in COVID-19 Symptom Time Series

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Abstract

Changes in population-level symptom reports may provide useful signals for public health surveillance, but identifying unusual reporting patterns is distinct from identifying viral variants. This study examines unsupervised anomaly detection in an hourly dataset containing eleven COVID-19-related symptom-reporting features from November 2020 through January 2021. Five baseline approaches are considered: principal component analysis (PCA), local outlier factor, vector autoregression, univariate long short-term memory (LSTM) models, and an LSTM autoencoder. A combined approach applies PCA, constructs overlapping multivariate windows, reconstructs those windows with a stacked LSTM autoencoder, and aggregates anomaly decisions across windows. The reported configuration retains eight components explaining approximately 93% of the variance and uses a 17-hour window. In the retained January 1–15 comparison, the combined model flags 41 hours under a mean-plus-one-standard-deviation threshold, compared with 19 for the baseline LSTM autoencoder. Under a mean-plus-two-standard-deviation threshold, both flag 18 hours. These results describe differences in alert volume and threshold sensitivity; they do not establish superior detection accuracy or successful prediction of SARS-CoV-2 variants. The study motivates further evaluation using independently validated events and temporally separated model selection.

Keywords: anomaly detection; multivariate time series; LSTM autoencoder; principal component analysis; syndromic surveillance.

1 Introduction

Symptom-reporting systems can complement conventional infectious-disease surveillance by recording changes in how people experience and report illness. Research on self-reported COVID-19 symptoms has demonstrated their potential for estimating probable infection, while also emphasizing the limitations of self-selected and self-reported observations [1]. At an aggregate level, an unusual combination of symptom counts may warrant investigation even when no labels identify the responsible event.

Anomaly detection offers a way to identify such departures from historical patterns. However, an anomalous hour can reflect a change in reporting participation, a data-quality problem, another circulating illness, or a change in disease activity. It cannot, by itself, establish the emergence of a viral mutation or variant. SARS-CoV-2 variant tracking relies on virological and epidemiological evidence, including genomic surveillance [2]. We therefore frame the task as exploratory detection of unusual symptom-reporting patterns rather than variant prediction.

This paper compares several statistical and neural approaches to that task and describes a PCA-assisted LSTM autoencoder. PCA provides a compact linear representation of the eleven input features. A sequence autoencoder then reconstructs overlapping windows of that representation, and a voting rule combines repeated decisions for each hour. The combination is motivated by the potential to capture temporal structure while limiting input dimensionality.

The contribution is an application-specific combination and explicit formulation of established methods, together with a descriptive comparison of retained experimental results. Sequence reconstruction with LSTM encoder-decoder networks predates this study [3]; neither multistep inputs nor three-dimensional input tensors are new capabilities of LSTMs. Our comparison concerns the particular configurations described here.

2 Related Work and Baseline Methods

2.1 Principal component analysis

PCA projects centered observations onto orthogonal directions ordered by explained variance [4]. As an anomaly detector, it can identify observations that are poorly reconstructed from a retained subspace. If $\mathbf{x}_t$ is a scaled observation and $\hat{\mathbf{x}}_t$ is its reconstruction in the same feature space, a possible score is

$$s_t^{\text{PCA}} = \|\mathbf{x}_t - \hat{\mathbf{x}}_t\|_2^2. \quad (1)$$

The comparison must use the inverse-transformed reconstruction, not subtract a lower-dimensional component vector directly from the original features. PCA is useful as an interpretable linear baseline, although retained variance is not a guarantee that all anomaly-relevant information has been preserved.

2.2 Local outlier factor

Local outlier factor (LOF) compares an observation’s local reachability density with the densities of neighboring observations [5]. Relatively low-density observations receive larger outlier scores. Neighborhood size determines the scale of the comparison: small neighborhoods emphasize local variation, whereas larger neighborhoods may smooth over locally unusual observations. A score greater than one indicates lower relative density, but it is not a universal decision threshold. LOF applied to hourly feature vectors does not explicitly model their temporal order.

2.3 Vector autoregression

A vector autoregressive model of order p expresses each feature using lagged values of all modeled features:

$$\mathbf{x}_t = \mathbf{c} + \sum_{\ell=1}^p \mathbf{A}_\ell \mathbf{x}_{t-\ell} + \boldsymbol{\varepsilon}_t. \quad (2)$$

Large forecast residuals can serve as anomaly scores. Standard stable VAR modeling requires appropriate treatment of nonstationarity and specification of lag order [6]. Strong contemporaneous correlation among every pair of features is not a prerequisite; predictive relationships may instead appear at particular lags. Seasonal structure and trends require explicit modeling or preprocessing.

2.4 Univariate LSTM models

LSTMs use gated recurrent states to learn temporal dependencies [7]. In the univariate baseline, a separate model is applied to each symptom series, and an hour is flagged globally when more than half of the eleven feature-level decisions are anomalous. This aggregation rule may suppress changes affecting only a small subset of symptoms. It is a modeling choice rather than an inherent property of LSTMs. Likewise, an LSTM does not inherently require Gaussian residuals; assumptions about residual distributions enter through a particular scoring or thresholding procedure.

2.5 LSTM autoencoders

An LSTM autoencoder maps an input sequence to a latent representation and reconstructs that sequence. Reconstruction error can indicate a departure from learned patterns [3]. This differs from forecasting an unseen next observation. The reported baseline uses a one-step input with the original features; the combined configuration uses PCA scores and a longer window. A longer window can expose temporal relationships, but whether it improves detection must be established empirically.

3 Data and Study Design

The study uses hourly observations from November 2020 through January 2021, with eleven COVID-19-related symptom-reporting features. November and December form the reported training period, and January forms the test period. The available study record does not provide a complete data dictionary, collection platform, geographic coverage, reporting denominator, or missing-data protocol. We therefore refer to the inputs as symptom-reporting features without assigning a more specific clinical interpretation to their counts.

Let $\mathbf{d}_t \in \mathbb{R}^{11}$ denote the observation at hour t . No independently verified anomaly labels or linked viral sequences are available in the supplied record. The outcome considered here is consequently the number of hours flagged by each method, rather than sensitivity, specificity, or variant-detection lead time.

The retained numerical comparison covers January 1–15, 2021, inclusive. Results were retained when their daily entries agreed with the corresponding reported subtotal. Contradictory entries and summaries were excluded. This restriction is a reporting decision, not a newly defined biological event window. The model outputs have not been regenerated from raw data for this revision.

4 PCA-Assisted Sequence Reconstruction

4.1 Preprocessing and dimensionality reduction

For an evaluation that avoids test-data leakage, the temporal split must precede fitting any data-dependent preprocessing. Feature scaling and PCA must be fitted on training observations and then applied unchanged to later observations [8]. Figure 1 specifies that ordering. This is the clarified evaluation protocol; the retained historical counts should not be interpreted as results from a new execution of it.

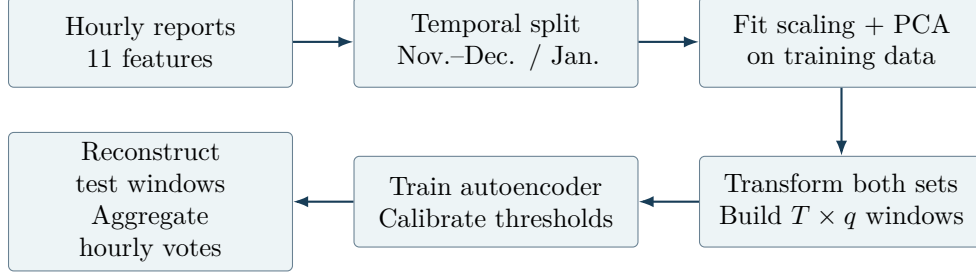


Figure 1: Clarified evaluation workflow. Preprocessing is fitted on training data, and threshold calibration precedes test scoring. The output is a set of unusual symptom-reporting hours requiring further investigation.

For feature j , min-max scaling can be written as

$$x_{t,j} = \frac{d_{t,j} - a_j}{b_j - a_j}, \quad a_j = \min_{t \in \mathcal{T}_{\text{train}}} d_{t,j}, \quad b_j = \max_{t \in \mathcal{T}_{\text{train}}} d_{t,j}. \quad (3)$$

Constant features require a documented convention, such as mapping them to zero. Test observations may lie outside $[0, 1]$ when they exceed training extrema; they should not be rescaled using test-specific bounds.

Let $\boldsymbol{\mu}$ be the training mean of the scaled features, and let $\mathbf{V}_q$ contain the first q eigenvectors of the training covariance matrix. The component scores are

$$\mathbf{z}_t = \mathbf{V}_q^T (\mathbf{x}_t - \boldsymbol{\mu}) \in \mathbb{R}^q. \quad (4)$$

With eigenvalues $\lambda_1 \geq \dots \geq \lambda_{11} \geq 0$, the retained dimension is

$$q = \min \left\{ r : \frac{\sum_{j=1}^r \lambda_j}{\sum_{j=1}^{11} \lambda_j} \geq 0.90 \right\}. \quad (5)$$

The reported configuration retains $q = 8$ components, accounting for approximately 93% of the variance. The first two components explain 22.9% and 19.8%, respectively. These are properties of the reported fit, not independently reproduced estimates. The retained components are linear combinations of features, not eight individual symptoms.

4.2 Overlapping windows and tensor dimensions

For a sequence of n component vectors, define a window of length T beginning at index i as

$$\mathbf{W}_i = \begin{bmatrix} \mathbf{z}_i^T \\ \mathbf{z}_{i+1}^T \\ \vdots \\ \mathbf{z}_{i+T-1}^T \end{bmatrix} \in \mathbb{R}^{T \times q}, \quad i = 1, \dots, n - T + 1. \quad (6)$$

Stacking these windows produces a tensor of shape $(n - T + 1, T, q)$. This reconstruction formulation uses every complete window and does not require a next-step target. A forecasting dataset that reserves an additional future observation would instead have $n - T$ examples. Training and test windows are constructed separately here.

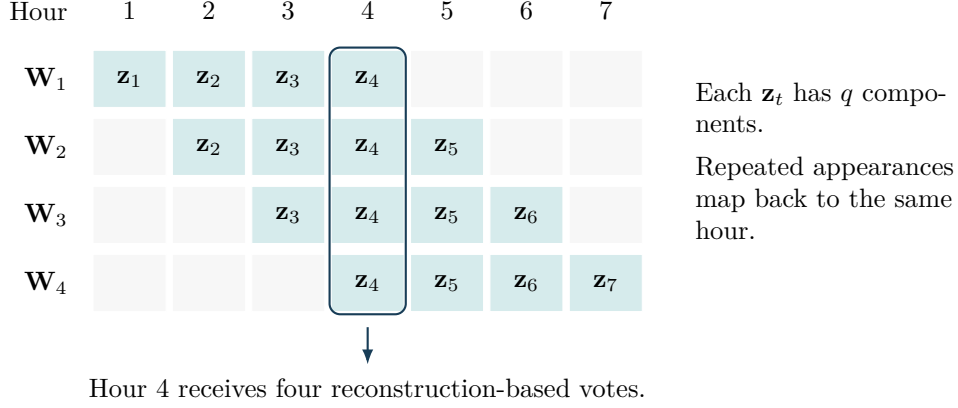


Figure 2: Window construction and overlap, illustrated with $T = 4$. Every row is a $T \times q$ matrix. The highlighted column shows repeated observations of the same hour; these repetitions are not new, independent time points.

4.3 Encoder–decoder architecture

The reported network contains two encoding LSTM layers with 60 and 30 units, followed by a repeat-vector operation and decoding LSTM layers with 30 and 60 units. A time-distributed dense layer maps each decoder state back to q components (Figure 3). The repeat-vector operation is not an LSTM layer. The supplied architecture specifies ReLU activations within the LSTM layers and a linear dense output.

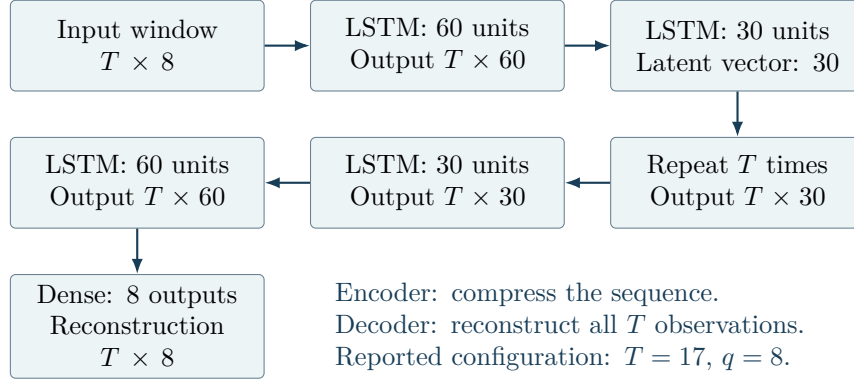


Figure 3: Editable schematic of the reported LSTM autoencoder. Dimensions exclude the batch axis. The dense output layer, rather than the final 60-unit recurrent layer, restores the eight-dimensional input representation.

Writing the network as f_θ , its output is $\widehat{\mathbf{W}}_i = f_\theta(\mathbf{W}_i)$. A reconstruction objective consistent with the reported error measure is

$$\mathcal{L}(\theta) = \frac{1}{mTq} \sum_{i=1}^m \sum_{k=1}^T \sum_{j=1}^q |W_{i,k,j} - \widehat{W}_{i,k,j}|, \quad (7)$$

where m is the number of training windows. The study reports 150 training epochs. Optimizer, learning rate, batch size, random seeds, and stopping criteria are not sufficiently specified to reproduce the reported run.

4.4 Window-length selection

The reported search starts at $T = 10$, increases or decreases the window length in one-hour increments, and compares the resulting training anomaly counts. The selected configuration uses $T = 17$. This is a reported heuristic selection, not evidence of a globally optimal window length. Maximizing the number of training alerts can favor an overly sensitive detector and does not estimate detection accuracy.

A stronger evaluation would select the window length on a later, temporally separated validation interval within the pre-test period. Candidate models should be compared using independently labeled events, where available, or a prespecified alert budget and stability criterion. January test results should remain excluded from model selection.

4.5 Thresholds and aggregation across windows

To make the scoring rule explicit, let the mean absolute reconstruction error for position k in window i be

$$e_{i,k} = \frac{1}{q} \sum_{j=1}^q |W_{i,k,j} - \widehat{W}_{i,k,j}|. \quad (8)$$

For the training errors, let μ_e and σ_e denote their mean and standard deviation. The two reported threshold rules are

$$\tau_1 = \mu_e + \sigma_e, \quad \tau_2 = \mu_e + 2\sigma_e. \quad (9)$$

Thresholds are model-specific: the same multiplier does not give different models the same false-alarm rate. These rules are heuristic cutoffs and should not be interpreted as calibrated statistical significance levels.

An occurrence of an hour is flagged when $e_{i,k} > \tau_a$, for threshold index $a \in \{1, 2\}$. Its original time index is $t = i + k - 1$. Let

$$\mathcal{I}_t = \{(i, k) : i + k - 1 = t\}, \quad c_t = |\mathcal{I}_t|. \quad (10)$$

An explicit majority decision is then

$$A_t^{(a)} = \mathbf{1} \left\{ \frac{1}{c_t} \sum_{(i,k) \in \mathcal{I}_t} \mathbf{1}\{e_{i,k} > \tau_a\} > \frac{1}{2} \right\}. \quad (11)$$

For an interior hour with $c_t = T = 17$, at least nine votes are required. Boundary hours appear in fewer windows, so the denominator must reflect their actual coverage; alternatively, evaluation can be restricted to fully covered hours. The historical results do not establish which boundary convention was implemented. Equation 11 makes the convention explicit for a reproducible implementation.

Overlapping windows also have an operational consequence. Some reconstructions of hour t use observations after t , and collecting all interior votes can require waiting as long as $T - 1$ hours. The reported approach is therefore a retrospective scoring method unless its detection delay is explicitly incorporated. An online version should score only completed windows and record the time at which each alert becomes available.

5 Retained Results and Discussion

5.1 Comparison over January 1–15

Table 1 presents the consistent retained counts. The PCA comparison is omitted because its original subtotal conflicts with its daily entries. LOF has a single reported decision setting, so it is not assigned an artificial second threshold. Figure 4 compares the four methods for which both threshold results are retained.

Table 1: Retained anomalous-hour counts for January 1–15, 2021. Counts are descriptive reported outputs, not numbers of confirmed mutations or validated detection events.

Method	Threshold 1	Threshold 2
Local outlier factor	7*	—
Vector autoregression	15	4
Univariate LSTM	13	7
LSTM autoencoder	19	18
PCA-assisted LSTM autoencoder	41	18

For the four paired methods, threshold 1 is $\mu_e + \sigma_e$ and threshold 2 is $\mu_e + 2\sigma_e$, using the respective model’s training-error statistics. *LOF uses its reported single setting; its count is not a mean-plus-standard-deviation comparison.

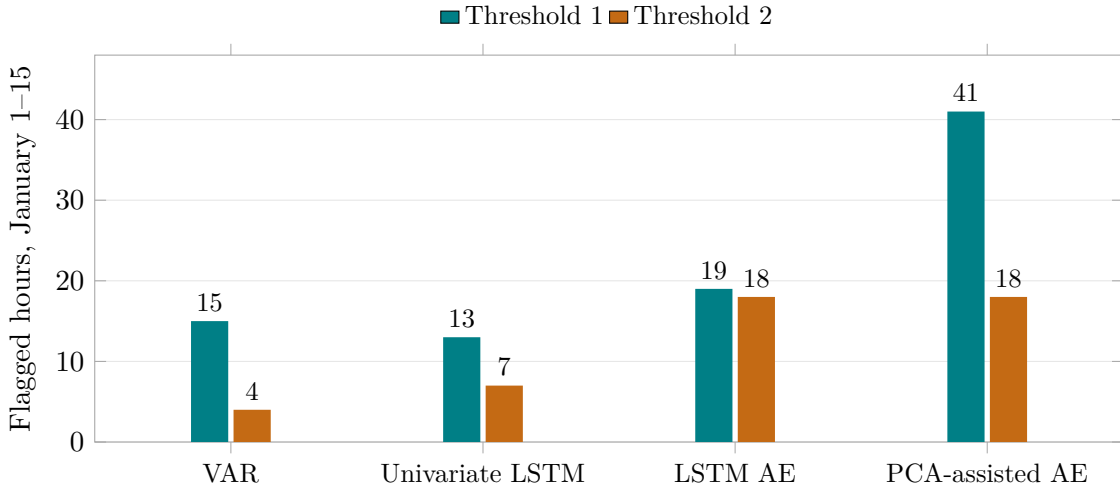


Figure 4: Threshold sensitivity in the retained comparison. The PCA-assisted autoencoder produces more alerts under threshold 1, but the two autoencoders have the same alert count under threshold 2. Higher bars indicate more flagged hours, not better detection accuracy.

Under threshold 1, the PCA-assisted autoencoder flags 41 hours, compared with 19 for the baseline autoencoder, 15 for VAR, and 13 for the univariate LSTM. The combined model thus generates 22 more alerts than the baseline autoencoder in this interval. Under threshold 2, both autoencoders flag 18 hours. Equal counts do not establish that the same hours were flagged, because timestamp-level agreement is not available in the retained summary.

The combined model’s count decreases by 23 when the threshold multiplier rises from one to two, whereas the baseline autoencoder’s count decreases by one. This indicates that the combined

model’s alert volume is more sensitive to the reported threshold change in this interval. It does not establish which setting is preferable: a larger alert volume might capture relevant events or increase false alarms.

5.2 What the comparison establishes

The results demonstrate that the selected representation, temporal window, and aggregation procedure yield a different alert pattern from the tested baselines. They do not isolate the contribution of PCA, window length, or voting, since those changes were introduced together. An ablation study would be needed to distinguish their effects.

The absence of verified events prevents estimates of precision, recall, or clinically meaningful lead time. Temporal proximity to a variant report is insufficient to establish a biological relationship, particularly when the symptom data and variant evidence are not linked by location or sampled population. Consequently, the retained counts support a comparison of alert volume rather than a ranking of mutation-prediction performance.

6 Limitations and Future Evaluation

The principal limitation is the absence of independently validated outcomes. A future study should link the symptom-reporting population to appropriately matched epidemiological and genomic observations and define relevant events before model fitting. Evaluation should report event-level sensitivity, false alerts per unit time, and detection delay, alongside uncertainty across repeated runs.

Data provenance is also essential. The eleven features require precise definitions, reporting denominators, geographic coverage, and a documented missing-data policy. Changes in platform participation or reporting behavior can shift hourly counts without a corresponding change in disease activity. Training data may themselves contain anomalies, which can influence both the learned representation and the estimated error thresholds.

A prospective evaluation should fit all preprocessing on past data, separate validation from testing, and specify how overlapping windows are handled at interval boundaries. It should account for the delay required to obtain repeated reconstructions. Thresholds should be calibrated against a common operational objective rather than assumed comparable because they use the same standard-deviation multiplier.

Finally, an ablation study should compare the original-feature autoencoder at $T = 1$ and $T = 17$, a PCA-assisted autoencoder at both window lengths, and aggregation with and without majority voting. Seasonal baselines and robust threshold estimators would help determine whether the additional neural-model complexity produces practical benefits. Further dimensionality-reduction methods or deeper networks should be evaluated only within the same controlled protocol.

7 Conclusion

This study describes a PCA-assisted LSTM autoencoder for exploratory anomaly detection in hourly COVID-19 symptom-reporting data. The method combines dimensionality reduction, overlapping sequence reconstruction, and aggregation of repeated hourly decisions. In the consistent

January 1–15 comparison, it produces more alerts than the baseline autoencoder under the less restrictive threshold, while both produce 18 alerts under the more restrictive threshold. The evidence supports a difference in alert volume and threshold sensitivity, not a claim of superior detection accuracy or successful variant prediction. Establishing surveillance value requires reproducible experiments, matched independent outcomes, and evaluation that accounts for false alarms and detection delay.

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